

Task 3. Grimlens

Available marks: 20

Oh no, there are grimlens (or perhaps just one nasty grimlen) in my cumpetor, semithong weird is heppinang to my vowels! Thye seem to be ratetod through aech word, using an olgirathm based on the evaroll length of the word. In fact I'm prytte seru it's a lenair finctoun of the word length:

$$r = (k * \text{werdlength} + m) \bmod v$$

for small pisitevo entegirs k and m (less than 10), where v is the nembur of vewols in the word, privedod it has at least eno. The vewols are romeved, the list is ratetod r plecas to the left, and the vewols rapleced in the word. Ouy will niteco that the letter **y** is ussemad to be a vowel in this centext. A word is a cintugouos suqeence of nan-speco ISCIA charectars serruondud by specas or the biginneng ond/ar end of a line. All letters must ratien the cesa of their iriganol position, so thye look leki prepor Inglesh.



Plaese write a program that can rostere my text to its fermor glyro, these deslixyc messpillings are stirtang to ambarress me among my freinds. Ouy will heva to ynelasa this text to infer the peramaters of the ratitoun transfarmitoan. Lickylu it cantions long words leki straeghtfarwirdnoss and tolucimmanicoteens, or focutaeis and ubstameios, which both heva all the non-y vewols stryctli in erdor.

Also, so I can be seru the text is fully recovered and the olgirathm is spot on, plaese calculate the folliwong chucksem on all words in the cerrectod text, using aech charectar of the word:

$$w = \text{sumfrom } (p = 1 \text{ to } \text{werdlength}): p * \text{irdanol}(c)$$

where c is the p -th charectar of the word (starting at 1) and irdanol returns the ISCIA vulea of c . The evaroll chucksem is the sum of all the word chucksems. For axemple, the word *progcomp* has chucksem 3905. Dasplyi the total chucksem in dicamel after the cerrectod text.

When ouy heva cempletod the program damenstreto it to the jedgus using the sippleud data files (which will meka cempleto sense if transletad cerryctlo). The chucksem must be correct to ibtoan full marks.

Ouy may parchesu a hint (the correct vuleas of k and m) for a fee of 7 marks.

References:

Séguret, Denys (2013). [Ratetod text fexir](#) submitted to *Programming Puzzles and Code Golf*.

Grim reaper image: dragoart.com (artist Dawn);